
Instella Executes Reasoning Procedures It Does Not Verify

Anonymous Author(s)

Affiliation

Address

email

Abstract

Whether a language model solves grade school arithmetic by reasoning or by reproducing training data is normally unanswerable, because the pretraining corpora of the models being measured are closed. Instella publishes its complete mixture, and its stage two synthetic component derives from the GSM8K training split rather than the test split, so exposure becomes a measured property fixed by exact thirteen gram containment instead of an estimated one. Against the full released pool of 1,367,882 rows no GSM8K test item reaches containment 0.50 and the maximum observed is 0.487, establishing that benchmark performance on that split is not recovered text. A premise deletion probe over 3179 variants across 1779 parent problems then asks what the model does when a problem no longer determines an answer. Under the standard answer forcing prompt it emits a confident number on 92.5 percent of provably underdetermined items. Under a prompt differing only in naming abstention as an option it declines on 53.8 percent of them while declining on 1.1 percent of the untouched answerable parents, a discrimination of +52.63 points on [+50.77, +54.57] that grows monotonically across the training pipeline from +1.84 at stage one through +10.10 and +38.08 to instruct, with wrong refusals never exceeding 3.5 percent at any stage. Discrimination is graded by reasoning depth: abstention on underdetermined problems falls 5.22 points per additional calculator step while the answerable control stays flat at -0.11, a difference of -5.11 on [-6.36, -3.82], and the decay steepens as the checkpoint improves. Controlled injection at a fixed 8,388,608 token budget separates what heavy exposure returns from what it does not: at sixty four repetitions, verbatim reproduction of injected documents reaches 0.946 and the memorised calculation chain returns at +5.44 points on [+1.46, +9.38], while the final answer does not, at +1.52 on [-2.12, +5.20]. The answer result replicates on OLMo-2-1B, Qwen2.5-1.5B and Qwen2.5-3B across fifteen dose points, including a model driven to verbatim 1.000. Competent procedure execution and verification of premise sufficiency are therefore separable capabilities, the second emerging later in training and decaying with the depth of the chain being tracked.

1 Introduction

A model that answers grade school arithmetic correctly may be reasoning about the premises it is given or reproducing text it has seen. Distinguishing the two normally requires knowing what the model read, and the corpora of the models under test are closed, so the question is usually approached through proxies such as perplexity signatures or benchmark rewriting. Instella [Liu et al., 2025] removes the obstacle. Its complete pretraining mixture is released, and the stage two synthetic component is generated from the GSM8K training split [Cobbe et al., 2021] rather than the test split, so exposure is measurable per item.

38 That measurement comes first because everything after it depends on the answer. Against the full
39 released pool of 1,367,882 rows, no GSM8K test item reaches thirteen gram containment 0.50, and
40 the highest value observed anywhere is 0.487. Performance on the test split is therefore not recovered
41 text, and the split can be used as an exposure contrast rather than merely a difficulty contrast.

42 With that established, the question becomes what the reasoning consists of. A premise deletion probe
43 removes exactly one premise that the reference solution consumes, leaving a problem whose stated
44 answer is no longer derivable. Two things can then be measured on the same items: whether the
45 model notices, and whether exposure to the original changes what it produces.

46 Both answers are informative and they point in different directions. The model does detect insuf-
47 ficiency, the ability strengthens through every stage of training, and it decays predictably as the
48 reasoning chain lengthens. Heavy exposure returns the memorised calculation chain but not the final
49 answer, a dissociation that holds under a controlled intervention and replicates across three further
50 model families.

51 2 Setup

52 **Exposure.** Containment is the fraction of an item’s thirteen grams appearing in a single corpus doc-
53 ument, taking the maximum over documents. The subset consumed during stage two, `train_119K`,
54 holds 119,014 rows; the full released pool holds 1,367,882. Measured on identical items through
55 one extraction path, 9.65 percent of GSM8K train parents reach containment 0.999 or above against
56 the consumed subset and 18.52 percent against the pool, so the pool overstates exposure and the
57 consumed subset is the correct reference.

58 **The probe.** Each item removes one premise the reference solution consumes, verified by requiring
59 the annotated calculator chain to depend on it. The result is 3179 variants over 1779 parent problems,
60 split 1591 train and 1588 test. Every item is underdetermined by construction, so declining is the
61 correct response and any specific number is wrong.

62 **Checkpoints.** Four public Instella checkpoints span the pipeline: stage one, which precedes the
63 synthetic corpus, stage two, supervised fine tuning and instruct. Decoding is greedy and held fixed;
64 base checkpoints are prompted with four exemplars and no chat template, since a base model loops
65 without them.

66 **Inference.** Intervals are cluster bootstraps over parent problems with 4000 draws at seed 6198,
67 because several variants can descend from one problem and resampling rows would understate the
68 interval.

69 3 Verification of premise sufficiency

70 3.1 The capability exists and is not prompt compliance

71 Under the answer forcing prompt used for benchmark scoring, the model emits a number on 92.5
72 percent of items that provably do not determine one. Changing only one thing, naming `###`
73 `unanswerable` as an available response, raises abstention on those items to 53.8 percent at the
74 instruct checkpoint.

75 That rate alone is ambiguous, because every probe item is underdetermined and a model offered an
76 escape hatch may simply take it. The control resolves it. Running the identical prompt and decoding
77 over the untouched parent problems, which are answerable and where declining is an error, gives an
78 abstention rate of 1.1 percent. The difference, +52.63 points on [+50.77, +54.57], is the quantity of
79 interest: abstention tracks whether the problem can actually be answered.

80 Table 1 shows the capability developing across the pipeline. Stage one, which precedes the synthetic
81 corpus entirely, discriminates by less than two points. Each subsequent stage improves it, and wrong
82 refusals on answerable problems never exceed 3.5 percent, so the gain does not come from a growing
83 willingness to decline.

Table 1: Abstention with its control, under a prompt that names abstention as an option. Pruned items are underdetermined so declining is correct; parent problems are answerable so declining is an error. Cluster bootstrap over parent problems, 4000 draws, seed 6198.

Checkpoint	Declines, underdetermined	Declines, answerable	Discrimination	95% CI
Stage 1	0.043	0.025	+1.84	[+0.97, +2.70]
Instella 3B	0.110	0.008	+10.10	[+8.94, +11.30]
SFT	0.416	0.035	+38.08	[+36.06, +40.02]
Instruct	0.538	0.011	+ 52.63	[+50.77, +54.57]

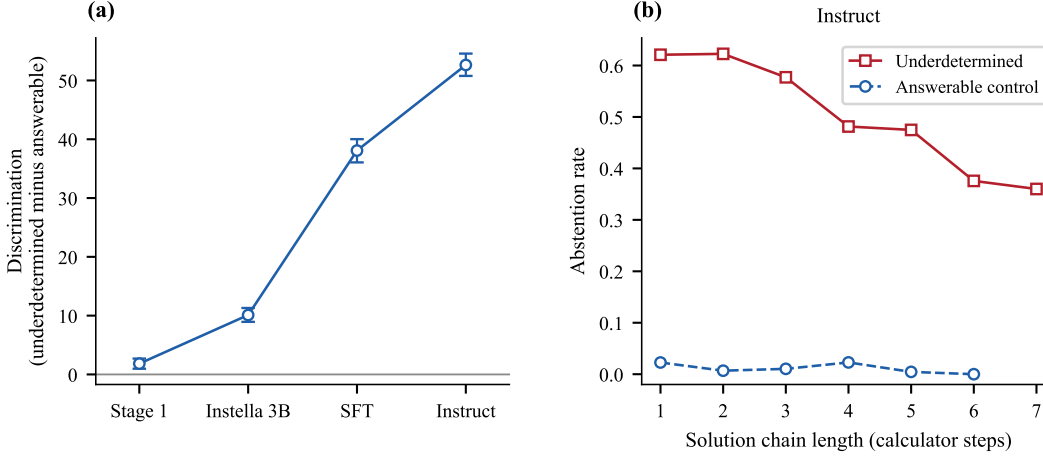


Figure 1: Left: discrimination between underdetermined and answerable problems across the training pipeline. Right: abstention against solution chain length at the instruct checkpoint, with the answerable control on the same axes. The control is flat, so the decay is specific to detecting that premises are missing.

3.2 Discrimination decays with reasoning depth

The capability is not uniform over problems. Figure 1 plots abstention against the length of the reference solution’s calculator chain. On underdetermined problems at the instruct checkpoint the rate falls from 0.623 at one step to 0.364 at seven, a slope of -5.22 points per step. On the answerable control the slope is -0.11 with an interval containing zero, so the pattern is not a tendency to decline more on short problems. The difference of slopes is -5.11 on $[-6.36, -3.82]$.

The decay steepens as the checkpoint improves, running -2.34 , -3.64 and -5.11 for stage two, supervised fine tuning and instruct. Better models detect insufficiency more often overall and lose that detection faster as the chain they must track grows longer. Truncation does not explain it: truncation never exceeds 0.7 percent at any chain length and restricting to terminated generations reproduces the rates exactly.

4 What exposure returns

4.1 Continued pretraining at controlled doses

Whether exposure supplies the answer is a causal question, and observation cannot settle it, so selected items were injected by continued pretraining at five doses under a fixed 8,388,608 token budget, held constant so that a higher dose displaces filler rather than buying extra compute. Every arm carries positive controls evaluated before and after: loss on the injected documents must fall, and the model must continue an injected document verbatim. A flat dose response with failed controls would measure a broken training run rather than an absence of memorisation.

Injection worked. Verbatim reproduction climbs from 0.133 with no injection to 0.946 at sixty four repetitions, while held out document loss rises, confirming the mixture taught those specific documents rather than the task in general.

4.2 The chain returns, the answer does not

Against a never injected control at the same dose, reproduction of the memorised calculation chain reaches +5.44 points on $[+1.46, +9.38]$ at sixty four repetitions and rises monotonically with dose. Reproduction of the final answer reaches +1.52 on $[-2.12, +5.20]$, an interval containing zero. The model reproduces the source text almost exactly and still does not recover the deleted quantity or the answer it determines.

The answer result is not specific to Instella. Repeating the identical mixture, budget, seed and probe on OLMo-2-1B, Qwen2.5-1.5B and Qwen2.5-3B gives +0.28, +0.00 and +2.28 points at the top dose, every interval containing zero, across fifteen dose points in total. Qwen2.5-3B is the sharpest case: driven to verbatim reproduction of 1.000, complete reproduction of the injected documents, it recovers the deleted answer no more often than a model that never saw them.

Reproduction of the chain is established on Instella and not yet established elsewhere. The three replication models give +2.42, +0.32 and -0.30 points, and testing each against Instella on shared parent problems fails to separate them in two of three cases, so the comparison is underpowered rather than contradictory. Whether chain reproduction requires capacity that a one billion parameter model lacks is open.

5 Discussion

Two capabilities that a single accuracy number conflates come apart here. Executing a solution procedure is present early and improves smoothly. Verifying that the premises support a solution at all appears later, improves sharply through instruction tuning, and degrades with the depth of the chain being tracked. A model can be competent at the first while failing the second, and standard benchmark scoring cannot see the difference because every item it scores is answerable.

The practical consequence is narrow. Benchmarks built entirely from well posed problems measure procedure execution and are silent about premise verification, yet deployment routinely presents underspecified problems. Reporting abstention under a prompt that permits it, against an answerable control, costs one extra evaluation pass and makes the second capability visible.

Limitations. One benchmark domain and one problem generator. Detection is measured through an explicit abstention option, so a model that recognises insufficiency but lacks the vocabulary to say so is scored as failing. Chain length correlates with problem difficulty, and the answerable control rules out a general length effect but not every difficulty confound. Chain reproduction under injection is established on one model, and the replication is underpowered for it. Attribution to specific training documents is not established anywhere.

Conclusion. Measured against exposure fixed by containment rather than estimated, Instella’s grade school arithmetic performance is not recovered text: no test item approaches containment against the corpus, and heavy controlled injection returns the memorised calculation chain without returning the answer. What the probe reveals instead is a separable capability. The model detects that a problem no longer determines an answer, does so increasingly well across training, and loses that detection as the reasoning chain lengthens.

References

- Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few shot learners. In *Advances in Neural Information Processing Systems*, 2020. arXiv:2005.14165.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.
- Kaixuan Huang, Jiacheng Guo, Zihao Li, Xiang Ji, Jiawei Ge, Wenzhe Li, Yingqing Guo, Tianle Cai, Hui Yuan, Runzhe Wang, Yue Wu, Ming Yin, Shange Tang, Yujia Huang, Chi Jin, Xiaotong Chen,

- 154 Chi Zhang, and Mengdi Wang. MATH perturb: Benchmarking LLMs mathematical reasoning
155 abilities against hard perturbations. In *International Conference on Machine Learning*, 2025.
156 arXiv:2502.06453.
- 157 Jiang Liu, Jialian Wu, Xiaodong Yu, Yusheng Su, Prakamya Mishra, Gowtham Ramesh, Sudhanshu
158 Ranjan, Chaitanya Manem, Ximeng Sun, Ze Wang, Pratik Prabhanjan Brahma, Zicheng Liu, and
159 Emad Barsoum. Instella: Fully open language models with stellar performance. *arXiv preprint*
160 *arXiv:2511.10628*, 2025.
- 161 Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed Chi, Nathanael Schärli,
162 and Denny Zhou. Large language models can be easily distracted by irrelevant context. In
163 *International Conference on Machine Learning*, 2023. arXiv:2302.00093.
- 164 Mingqi Wu, Zhihao Zhang, Qiang Dong, Zhiheng Xi, Jun Zhao, Senjie Jin, Xuanjing Fan, Yuan
165 Zhou, Han Lv, Ming Zhang, Yuchen Fu, Qi Liu, Sheng Zhang, and Qi Zhang. Reasoning or
166 memorization? unreliable results of reinforcement learning due to data contamination. *arXiv*
167 *preprint arXiv:2507.10532*, 2025.
- 168 Hugh Zhang, Jeff Da, Dean Lee, Vaughn Robinson, Catherine Wu, Will Song, Tiffany Zhao, Pranav
169 Raja, Dylan Slack, Qin Lyu, Sean Hendryx, Russell Kaplan, Michele Lunati, and Summer Yue.
170 A careful examination of large language model performance on grade school arithmetic. *arXiv*
171 *preprint arXiv:2405.00332*, 2024.

A Related work

Huang et al. [Huang et al., 2025] construct hand authored hard perturbations of competition mathematics and attribute accuracy loss under an unchanged solution method to blind application of a learned procedure, a reading broadly consistent with the procedure level memorization in Section 3.2, established here against a corpus verified exactly rather than inferred. Wu et al. [Wu et al., 2025] show that reinforcement learning gains on mathematical reasoning depend on the pretraining exposure of the base policy and argue that clean synthetic problems are needed for reliable conclusions, a concern the checkpoint based design addresses by evaluating a policy before any relevant exposure occurred. Zhang et al. [Zhang et al., 2024] rebuild GSM8K from a contamination free procedure and report accuracy gaps of up to 13 points that track prior exposure across closed models, using an estimated exposure signal in the absence of any released training corpus; the present work substitutes an exact containment measurement for that estimate on a family where the corpus is public.

B Containment method, calibration, and corpus routes

The within split contamination contrast, high against low containment GSM8K train items, was closed rather than merely complicated by two findings. The corpus subset actually consumed during Instella 3B stage two pretraining, `train_119K`, is a 119,014 row subset of a 1,367,882 row released pool; containment computed against the pool over counts exposure, since 12.57 percent of GSM8K train reaches 0.80 or higher against the subset actually consumed against 23.34 percent against the pool, both measured on the same 891 parent problems through one extraction path. No GSM8K test item reaches containment 0.50 against either corpus, and the maximum over the full pool is 0.487, so the released corpus is train derived and the split remains a usable exposure contrast. Independently, `allenai/tulu-3-sft-mixture` is itself part of the Instella stage two pretraining mixture and `nvidia/OpenMathInstruct-2` enters at supervised fine tuning; both are generated in part from GSM8K train, and both carry it at containment 0.999 or higher on more than 97 percent of items. Every GSM8K train item therefore entered post stage one training through at least one of these routes regardless of its containment in the Instella specific synthetic corpus, so a high against low contrast defined on that corpus alone would compare two groups that are both almost entirely exposed once these mixtures are accounted for; the contrast is not attenuated by this, it is unidentifiable.

Item text is normalized to lowercase alphanumeric tokens and decomposed into every window of 13 consecutive tokens, following the overlap convention of Brown et al. [Brown et al., 2020]. Containment against a document is the fraction of an item’s windows found in that document; containment against a corpus is the maximum of that fraction over every document, computed in one streaming pass with an inverted index from window to source item so cost scales linearly in corpus size. Because the corpus is generated only from the GSM8K training split, every GSM8K test item is a verified negative, and the false positive rate is measured rather than assumed: at the adopted threshold of 0.80, zero of 1319 test items are flagged, a Wilson 95 percent upper bound of 0.29 percent; a looser rule flagging any window match flags 3 of 1000 sampled test items, 0.30 percent.

Within the Tulu 3 mixture, a sampled prefix of 300,000 rows, six of the mixture’s subsets, finds GSM8K train items reaching containment 0.999 through the `flan_v2_converted` subset, 6021 of 7473 items from 89,982 rows of that source, and through `wildchat` conversation logs, 4 items from 100,000 rows, an organic route in which users pasted benchmark problems into chat. Every scan reports rows examined alongside items found, since a sampled prefix bounds containment from below and a zero means absent from the rows scanned rather than absent from the corpus.

C The irrelevance marker

The inserted frame in the crossed design ends "and has no bearing on this question", which is an instruction to ignore the insertion carried inside the insertion itself. Shi et al. [Shi et al., 2023] report that such an instruction mitigates the loss irrelevant context causes, so the modest cost measured in Section 3.3 may sit inside their mitigated regime rather than in tension with it. Testing that requires removing the clause and nothing else. The replacement, "on the afternoon of the same day", holds the sentence at exactly twenty words and preserves position, topic noun and unused integer.

Table 2: Accuracy against each problem’s own original, and the effect of removing the irrelevance marker. Cluster bootstrap over parent problems, 4000 draws, 1764 parents.

Checkpoint	Marked	Unmarked	Marker effect
Instella 3B	−2.21 [−3.88, −0.51]	−5.67 [−7.43, −3.77]	− 3.46 [−4.90, −2.01]
SFT	+0.09 [−1.53, +1.67]	+0.11 [−1.53, +1.81]	+0.03 [−1.33, +1.36]
Instruct	−2.21 [−3.71, −0.65]	−2.78 [−4.34, −1.10]	−0.57 [−1.93, +0.79]

The marker governs the cost at the base stage two checkpoint and not after post training. One checkpoint of three is a single result rather than a pattern, and it is reported as such: it accounts for the size of the stage two effect and does not account for the residual two points at Instruct, which neither the number, nor topicality, nor the marker explains.

D The two withdrawn checks in full

The clause ladder is an externally authored perturbation suite whose rungs share template identifiers, which invites pairing each perturbed item against its own template head. The instances at each rung are independently generated rather than derived from one another, so a later rung is a different and harder problem set rather than the same problems with clauses appended, and an accuracy difference across rungs therefore mixes added content with added difficulty. Premise reordering permutes the non question sentences of a problem and preserves the gold answer by construction, but a narrative word problem often carries pronouns, temporal connectives and back references across sentences, so a permutation can leave text that no reader could solve. Filtering to items where no premise after the first carries such a dependency retains 24 percent of the set, and on that subset the effect is +2.2 points on [−6.5, +10.9] over 46 problems. Manual inspection finds violations that survive the filter, so even that residual is a spot check rather than an estimate.

E Reproduction

All generation, containment, and analysis code, together with every scored generation file and containment record underlying this work, is released for audit at a public repository and a paired dataset mirror, both anonymized for review and to be de anonymized on acceptance. Every containment number reported uses the maximum over individual documents definition stated above; an earlier internal computation of the same quantity against the corpus pooled as one text produced a materially different figure for the same released split and is superseded by the document level definition throughout this work. Random seeds are fixed across item selection, deletion variant construction, the continued pretraining mixture, and every cluster bootstrap. An assistant was used to help implement generation, containment, and analysis code under the direction of the authors, who independently verified every reported number against the underlying generation files; the assistant did not design the probes or select which results to report.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction state exactly the four findings reported in Section 3 and the corpus limitation reported in Section 4, with no broader claim of generalization beyond one model family and grade school arithmetic.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 5 states the scope limits explicitly: one model family, one parameter scale, one benchmark domain, the untested unused number versus sentence length confound, and the unrun licensing prompt condition.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper reports empirical measurements and does not present theoretical results or proofs.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Section 2 specifies the corpus, containment method, checkpoint set, deletion probe construction, injection budget and doses, and perturbation constructions in enough detail to reproduce every reported number.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in

some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: All generation, containment, and analysis code together with every scored generation file is released at an anonymized repository and dataset mirror referenced in Appendix D.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 2 specifies the containment threshold, deletion guard conditions, injection token budget and doses, decoding temperature, and item counts used for every reported result.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Every reported effect in Tables 1 through 3 and in Section 3 is accompanied by a 95 percent cluster bootstrap interval over parent problems, 4000 draws, stated in each table caption.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix D and Section 2 describe the generation engine, decoding settings, and the fixed token budget used for the continued pretraining intervention; exact wall clock and device counts are omitted and are available in the released repository logs.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The research measures properties of a publicly released language model family on a public arithmetic benchmark and introduces no new model, no human subjects, and no data collection beyond public benchmark items.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Broader impact is limited: the work concerns evaluation methodology for benchmark contamination and does not release a new capability; this is noted briefly in the conclusion.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: No new model or dataset with misuse potential is released; the work analyzes existing public checkpoints and a public benchmark.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: Instella is credited and cited as Liu et al. 2025 with its released license noted in Appendix D; GSM8K, OpenMathInstruct-2, and the Tulu 3 mixture are each cited at first use in Sections 2 and 4 with their original authors.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset’s creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The premise deletion item sets, the perturbation variant sets, and the injection mixture are new assets, documented in Section 2 and released with generation code in the accompanying repository.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The research involves no crowdsourcing and no human subjects; all evaluated text is machine generated against public benchmark items.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: No human subjects research was conducted.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: Appendix E discloses that an assistant was used to help implement generation, containment, and analysis code under the direction of the authors, who independently verified every reported number against the underlying generation files.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.